

Network Operating System

Unit 4

Multi Tier Computing & DCE

1 Distributed Computing Environment

1.1 Overview

The Distributed Computing Environment (DCE) from the Open Software Foundation (OSF) creates an open Enterprise Network Operating System environment that spans multiple architectures, protocols, and operating systems.

DCE provides several key distributed technologies including:

- Remote Procedure Call (RPC)
- Distributed Naming Service
- Time Synchronization Service
- Distributed File System
- Network Security Service
- Threads Package

DCE allows a client to interoperate with one or more server processes on different computing platforms, even when they come from different vendors and run different operating systems.

It also provides an integrated approach to:

- Security
- Naming
- Interprocess Communication

2 DCE Remote Procedure Call (RPC)

The DCE RPC initially originated from HP.

DCE provides an **Interface Definition Language (IDL)** and compiler to facilitate the creation of RPCs.

The IDL compiler creates portable C code stubs for both the client and server sides of an application.

These stubs are compiled and linked to the RPC runtime library which is responsible for:

- Finding servers in a distributed system
- Performing message exchanges
- Packing and unpacking message parameters
- Processing errors that occur

DCE RPC can be integrated with the DCE security and naming services. This allows:

- Authentication of each procedure call
- Dynamic location of servers at runtime

Servers can concurrently service RPCs using threads.

The RPC mechanism also provides:

- Protocol independence
- Network independence

3 DCE Distributed Naming Services

The naming service is based on the **X.500 standard**.

DCE naming services allow resources such as:

- Programs
- Servers
- Files
- Disks
- Print queues

to be identified by user-oriented names stored in a distributed database.

Object names are independent of their location on the network.

3.1 Cells

DCE divides the distributed environment into administrative units called **cells**.

- A cell contains both client and server workstations.
- The cell domain is defined by the organization.
- Typically contains machines used by groups working on related tasks.
- Cell size depends on administrative convenience.

At minimum, a DCE cell must have:

- One cell directory server
- One security server

4 DCE Distributed Time Service

The DCE Time Service synchronizes computers in the network to a recognized time standard.

It provides APIs for:

- Manipulating timestamps
- Obtaining universal time

Public sources such as the **Traconex/PSTI radio clock** can be used.

Requirements:

- At least three Time Servers
- At least one connected to an External Time Provider

Time servers periodically query each other to adjust clocks.

DCE uses the **UTC standard**, which measures time elapsed since the beginning of the Gregorian calendar (October 15, 1582).

5 DCE Distributed Security Services

DCE security is based on MIT's **Kerberos authentication system** with enhancements from HP.

The Kerberos system includes:

- Authentication Server
- Security Database
- Privilege Server

These components reside on a secure server.

DCE network security services provide:

- Authentication
- Authorization
- User account management

The security server stores sensitive data such as:

- User names
- Passwords

The RPC mechanism hides most security complexity from users by:

- Obtaining authentication tickets
- Providing encryption
- Performing authenticated checksums

Authorization is implemented using **Access Control Lists (ACLs)**.

Each implementation must provide an ACL manager to control access to services and resources.

6 Distributed File System (DFS)

The DCE Distributed File System provides:

- Uniform namespace
- File location transparency
- High availability

DFS is log-based, which enables fast restart and recovery after server crashes.

Features include:

- Replication of files and directories across multiple servers
- Cache consistency protocol
- Automatic propagation of changes across caches

DFS provides a **single-image file system** distributed across multiple file servers.

7 Threads

DCE uses the **Concert Multithread Architecture (CMA)** from Digital.

Characteristics:

- Portable thread package
- Runs in user space
- Uses wrapper routines to translate calls to native kernel threads

Threads are essential for:

- Client-server applications
- Supporting other DCE components

8 Enterprise Network Operating System (NOS)

An Enterprise NOS manages:

- Network resources
- Users
- Security
- Services across an organization

8.1 Examples

- Windows Server
- Linux Server (Ubuntu Server, Red Hat)
- UNIX Systems

8.2 Features

- Centralized user management
- Resource sharing (files, printers)
- Security and access control
- Network monitoring
- Backup and recovery
- Scalability

8.3 Functions

- Authentication and authorization
- Directory services (LDAP / Active Directory)
- Distributed file systems
- Communication services
- Remote administration
- Load balancing

9 Internet as a Network Operating System

The Internet functions as a global NOS by providing shared services across distributed systems.

Capabilities include:

- Communication infrastructure
- Resource access
- Distributed applications
- Global connectivity

9.1 Services

- Email services
- Web services (HTTP)
- Cloud computing
- Remote login (SSH)
- File transfer (FTP)
- DNS name resolution

10 Secure Sockets Layer (SSL)

SSL provides secure communication between web browsers and servers.

Features:

- Authentication
- Encryption
- Message integrity

SSL operates between TCP/IP transport and sockets.

Web servers using SSL operate on **port 443** instead of port 80.

SSL provides:

- Private client-server communication
- Server authentication
- Reliable message integrity checks

Before data transmission, the **SSL Handshake Protocol** occurs. This process allows:

- Authentication
- Encryption algorithm negotiation
- Generation of shared secrets

SSL uses public-key cryptography developed by RSA Data Security.

11 Secure HTTP (S-HTTP)

S-HTTP is a security-enhanced version of HTTP developed by EIT.

It adds application-level encryption over standard HTTP communications.

Security features include:

- Client and server authentication
- Certificate revocation checks
- Certificate chaining
- Digital signatures
- Negotiation of security levels
- Secure communication through corporate firewalls

12 Self Study

Thin Client topics:

- Management and Scalability
- Access
- Performance
- Security
- Total Cost of Ownership